Healthcare Admissions, Outcomes, and Billing Dashboard

Power BI Development Report

Prepared by Eric Benitez and William Schmiedeknecht

Career Simulation Project - Power BI Portion

This report documents the dashboard design, DAX measures, calculated fields, visual development, interactivity, validation, and main findings from the Power BI portion of the healthcare analytics project.

Executive Summary

The Power BI portion of this project transformed an analysis-ready healthcare CSV into a two-page interactive dashboard. The dashboard was designed for a hospital leadership audience and focuses on admissions activity, patient demographics, clinical outcomes, length of stay, and billing patterns.

The dataset contained 55,500 patient encounters with admission dates from May 08, 2019 through May 07, 2024. Key Power BI measures showed total billing of $1,211,844,977, average billing of $21,835, an average length of stay of 17.84 days, an abnormal test rate of 50.06%, and an emergency or urgent admission rate of 57.51%.

The main dashboard finding was that long-stay clinical conditions also carried higher average billing amounts. Alzheimer's, Cancer, and Heart Disease produced the longest average stays and the highest average billing levels, making them important focus areas for capacity planning, care coordination, and financial review.

Dashboard Deliverables

Deliverable	Status	Notes
Admissions Overview page	Completed	Executive KPI cards, admissions trend, admission type, age group, hospital volume, and page-level slicers.
Clinical and Financial Analysis page	Completed	Gender, test results, insurance billing, length of stay, average billing matrix, KPI cards, and interactive slicers.
Core DAX measures	Completed	Admissions, billing, average stay, abnormal test rate, and emergency or urgent rate.
Calculated fields	Completed	Age Group and Age Group Sort columns for demographic analysis.
Interactivity testing	Completed	Slicers were tested and verified to update visuals as expected.

Dataset Scope Used in Power BI

Dataset item	Power BI use
Source file	healthcare_analysis_ready_revised.csv
Rows	55,500 encounters
Date range	May 08, 2019 to May 07, 2024
Primary key used	encounter_id
Financial fields	billing_amount and billing_per_day
Clinical fields	medical_condition, admission_type, medication, test_results, length_of_stay
Demographic fields	age, gender, blood_type
Operational fields	hospital, insurance_provider, admission_date, discharge_date, admission_month

Business Questions Answered by the Dashboard

#	Business question	Dashboard element	Answer / insight
1	How many total patient admissions occurred?	Total Admissions KPI	55,500 admissions
2	How have admissions changed over time?	Admissions Trend Over Time	Monthly admissions remained generally stable after removing incomplete edge months.
3	What percentage of admissions were Emergency, Elective, Urgent, or Routine?	Admissions by Type donut chart	Emergency was the largest category at 36.22%; Emergency/Urgent combined were 57.51%.
4	Which hospitals handled the greatest number of admissions?	Admissions by Hospital bar chart	Northwestern Memorial Hospital led slightly, but all four hospitals were close in volume.
5	Which age groups account for the most admissions?	Admissions by Age Group column chart	The 50-64 age group had the highest count at approximately 14.0K admissions.

6	How does admission volume differ by gender?	Admissions by Gender donut chart	Gender was balanced: Female 50.37%, Male 49.63%.
7	What percentage of test results were Normal, Abnormal, or Inconclusive?	Test Result Distribution donut chart	Abnormal 50.06%, Normal 31.45%, Inconclusive 18.49%.
8	Which medical conditions have the longest average length of stay?	Average Length of Stay by Condition bar chart	Alzheimer's was highest at about 54 days, followed by Cancer and Heart Disease.
9	Which insurance providers account for the greatest billing volume and highest average bills?	Billing by Insurance Provider matrix	Cigna had the highest total billing; Aetna had the highest average billing.
10	Which conditions and admission types generate the highest average billing amounts?	Average Billing by Condition and Admission Type matrix	Cancer, Heart Disease, and Alzheimer's showed the highest average billing levels.

Power BI Development Process

The Power BI workflow followed a structured build process: import the CSV, verify and set data types, create core measures, build calculated fields, design the dashboard pages, add visuals, apply formatting, test slicers, and validate the results against expected totals.

1. Data Import and Preparation

- Imported the CSV file into Power BI Desktop using Get Data > Text/CSV.
- Selected Transform Data rather than loading immediately so the table could be reviewed in Power Query.
- Renamed the query to Healthcare to make measures and formulas easier to read.
- Confirmed that the file contained 19 columns and 55,500 encounter records.
- Set identifier and category fields to Text, date fields to Date, count/flag fields to Whole Number, and billing fields to Decimal Number.

Data type	Columns
Text	encounter_id, gender, blood_type, medical_condition, hospital, insurance_provider, admission_type, medication_source, medication, test_results
Date	admission_date, admission_month, discharge_date
Whole number	age, admission_year, length_of_stay, abnormal_test_flag
Decimal number	billing_amount, billing_per_day

2. Core Measures Created

Measures were used instead of raw column aggregations so the dashboard could apply consistent logic across all visuals and slicers.

Measure	DAX logic
Total Admissions	DISTINCTCOUNT('Healthcare'[encounter_id])
Total Billing	SUM('Healthcare'[billing_amount])
Average Billing	AVERAGE('Healthcare'[billing_amount])
Average Length of Stay	AVERAGE('Healthcare'[length_of_stay])
Abnormal Test Count	SUM('Healthcare'[abnormal_test_flag])
Abnormal Test Rate	DIVIDE([Abnormal Test Count], [Total Admissions])
Emergency or Urgent Admissions	CALCULATE([Total Admissions], 'Healthcare'[admission_type] IN { "Emergency", "Urgent" })
Emergency or Urgent Rate	DIVIDE([Emergency or Urgent Admissions], [Total Admissions])

3. Calculated Columns Created

Two calculated columns were added to support age-group analysis and correct visual sorting. The initial sorting column created a circular dependency because it referenced Age Group while Age Group was being sorted by that same field. This was corrected by making the sort column depend directly on the numeric age field.

```
Age Group = SWITCH(TRUE(), age < 18, "0-17", age <= 34, "18-34", age <= 49, "35-49", age <= 64, "50-64", age <= 79, "65-79", "80+")
Age Group Sort = SWITCH(TRUE(), age < 18, 1, age <= 34, 2, age <= 49, 3, age <= 64, 4, age <= 79, 5, 6)
```

After both fields were created, Age Group was sorted by Age Group Sort so the chart displayed age bands in a logical order instead of alphabetical order.

4. Dashboard Page Design

Page	Purpose	Main visuals
Admissions Overview	Executive admissions summary	KPI cards, monthly trend, admission type share, age group distribution, hospital volume, admission date slicer, hospital slicer.
Clinical and Financial Analysis	Clinical outcomes, utilization, and billing	Gender split, test result distribution, emergency or urgent rate, average stay, average billing, length of stay by condition, billing by insurer, billing by condition and admission type, interactive slicers.

5. Visual and Formatting Choices

- Used KPI cards for high-level executive metrics at the top of the dashboard.
- Used a line chart for admissions over time because month-by-month order matters.
- Filtered the admissions trend to complete months only, using June 2019 through April 2024, to avoid misleading partial-month drops at the beginning or end of the date range.
- Used donut charts for small part-to-whole categories such as admission type, gender, and test results.
- Changed the initial condition-count bar chart into Average Length of Stay by Condition because condition counts were too evenly distributed to provide a meaningful visual story.
- Used matrices for insurance and billing analysis so total admissions, total billing, and average billing could be compared together.
- Applied currency formatting to financial measures to keep billing visuals consistent.
- Applied conditional background formatting to the billing matrix so high-value condition/admission combinations were easier to identify.
- Added dropdown slicers for interactivity and tested them to confirm visuals updated correctly.

Dashboard Screenshots

The final Power BI dashboard contains two pages. The first page provides an executive admissions overview, while the second page focuses on clinical, operational, and billing patterns.

Figure 1. Admissions Overview dashboard page

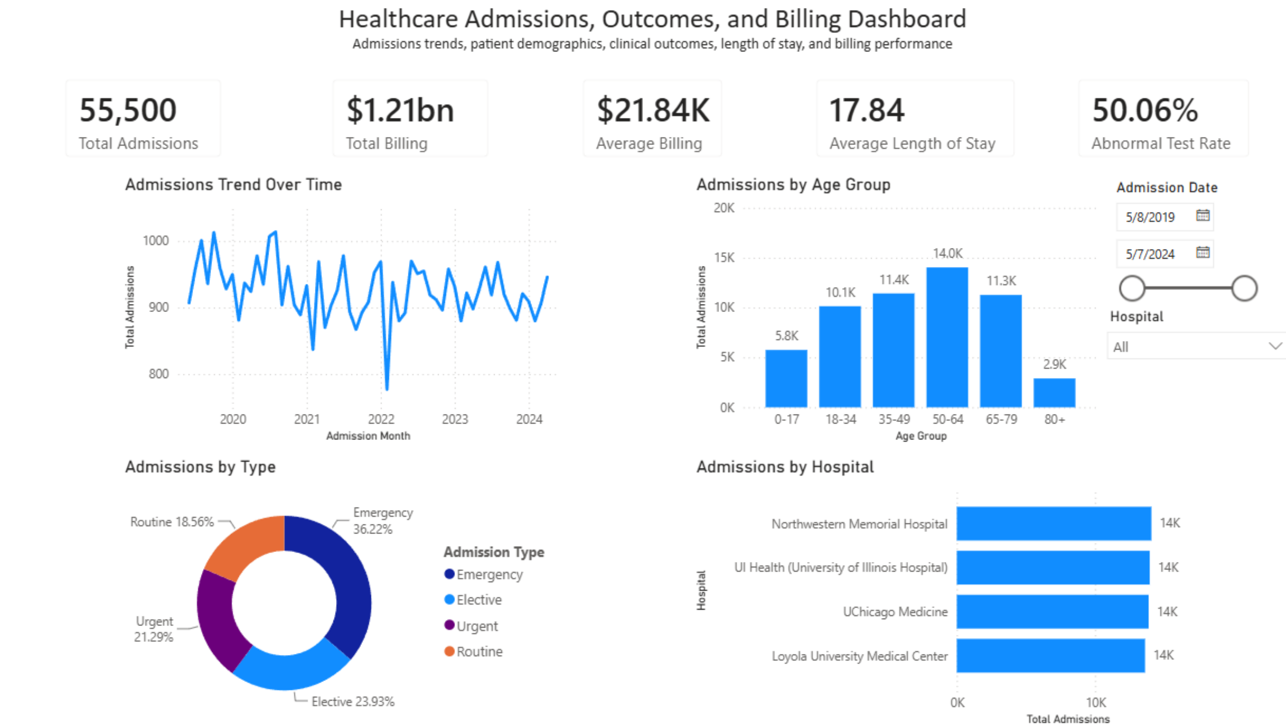
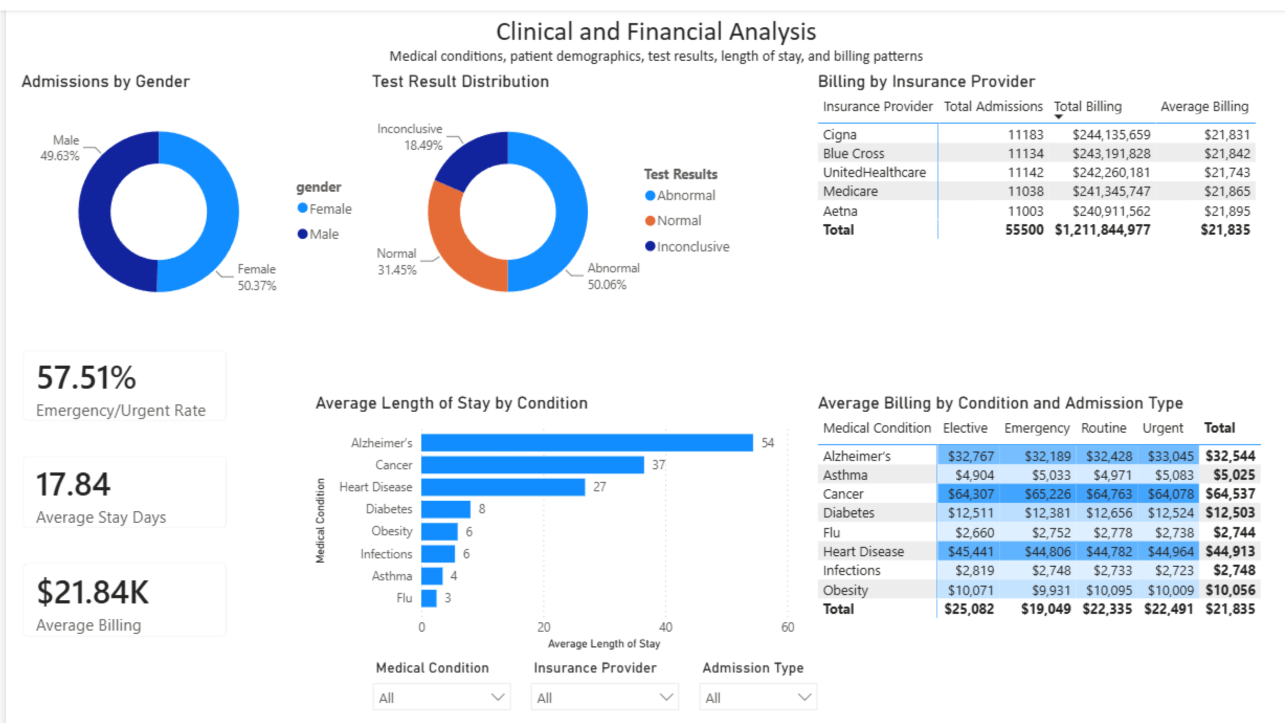


Figure 2. Clinical and Financial Analysis dashboard page



Insurance Provider	Total Admissions	Total Billing	Average Billing
Cigna	11183	$244,135,659	$21,831
Blue Cross	11134	$243,191,828	$21,842
UnitedHealthcare	11142	$242,260,181	$21,743
Medicare	11038	$241,345,747	$21,865
Aetna	11003	$240,911,562	$21,895
Total	**55500**	**$1,211,844,977**	**$21,835**

Medical Condition	Elective	Emergency	Routine	Urgent	Total
Alzheimer's	$32,767	$32,189	$32,428	$33,045	**$32,544**
Asthma	$4,904	$5,033	$4,971	$5,083	**$5,025**
Cancer	$64,307	$65,226	$64,763	$64,078	**$64,537**
Diabetes	$12,511	$12,381	$12,656	$12,524	**$12,503**
Flu	$2,660	$2,752	$2,778	$2,738	**$2,744**
Heart Disease	$45,441	$44,806	$44,782	$44,964	**$44,913**
Infections	$2,819	$2,748	$2,733	$2,723	**$2,748**
Obesity	$10,071	$9,931	$10,095	$10,009	**$10,056**
Total	**$25,082**	**$19,049**	**$22,335**	**$22,491**	**$21,835**

Key Findings from the Dashboard

Emergency demand is the largest admission category. Emergency admissions accounted for 36.22% of encounters, and Emergency plus Urgent admissions accounted for 57.51%.

Admissions volume is highest among middle-aged and older adults. The 50-64 age group had the highest admission count at about 14.0K encounters, followed by 35-49 and 65-79.

Gender distribution is balanced. Admissions were nearly evenly split between female and male patients, so gender did not appear to be a major volume driver in this dataset.

Abnormal test results are common. The abnormal test rate was 50.06%, with Normal results at 31.45% and Inconclusive results at 18.49%.

Length of stay varies strongly by condition. Alzheimer's averaged about 54 days, Cancer averaged about 37 days, and Heart Disease averaged about 27 days, far above short-stay conditions such as Flu and Asthma.

Billing is strongly tied to condition severity and stay length. Cancer, Heart Disease, and Alzheimer's produced the highest average billing levels, creating a clear link between clinical mix, capacity, and financial impact.

Insurance billing volume is evenly distributed. Total billing by insurer ranged from approximately $240.9M to $244.1M, with Cigna highest in total billing and Aetna highest in average billing.

Recommendations

- Use long-stay conditions such as Alzheimer's, Cancer, and Heart Disease as focus areas for capacity planning and bed-utilization review.
- Monitor Emergency and Urgent admissions closely because they represent more than half of total encounters and may require flexible staffing and resource planning.
- Use the billing matrix to review high-cost condition/admission combinations and identify where case management or financial review may be most valuable.
- Track abnormal and inconclusive test-result rates over time because these outcomes may influence follow-up workload, clinical review, and patient throughput.
- Continue using slicers by medical condition, insurance provider, admission type, hospital, and admission date so stakeholders can move from executive overview to targeted analysis.

Limitations

- The dataset appears to be synthetic and should be treated as a training dataset rather than real patient or hospital performance data.
- Billing amount was treated as a currency field for dashboard readability, but the source data does not provide reimbursement, payment, or cost-accounting detail.
- The dashboard analyzes encounter-level rows. It does not distinguish repeat admissions by the same real patient unless a reliable patient identifier is available.
- The admissions trend chart uses complete months only to avoid misleading partial-month results, while the KPI cards reflect the full dataset.
- The dataset does not include doctor/provider fields, so provider workload analysis was excluded from the final dashboard.

Power BI Contribution Summary

My portion of the project focused on building the Power BI dashboard layer from the analysis-ready healthcare dataset. This included importing and preparing the file in Power Query, setting data types, creating calculated fields and DAX measures, designing two dashboard pages, selecting appropriate visuals, applying formatting and conditional formatting, adding slicers for interactivity, validating that filters worked, and reviewing the dashboard for readability and business usefulness.

The final dashboard answers the required ten business questions and provides an executive-level view of admissions, demographics, clinical outcomes, length of stay, and billing performance. The project also demonstrates the ability to move beyond simply creating charts by evaluating whether a visual adds meaningful insight, revising weak visuals, and choosing stronger analyses when the data supports them.